MEA OCPP 1.6 Compliance Test Report

Section 9: MEA-Specific Configuration & V2G/BPT

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

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1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
MeterValues Timeout	15 s per power-demand item
Test Date	2026-06-10 08:21:20 UTC
Results file	tex/vsecc_section9_results.json

Test Architecture

The test calls the MEA sandbox REST API with HTTP Digest authentication to issue `ChangeConfiguration` commands, then subscribes to the vSECC MQTT broker to observe `metervalues` publications as confirmation that power-demand setpoints were received by the charger.

MEA-Specific OCPP Keys

The MEA CSMS uses two non-standard OCPP 1.6 configuration keys for V2G/BPT control:

- **V2GMode** — Enables bidirectional power transfer (BPT) capability at the CSMS level. Value: `"true"` or `"false"`.
- **Power.Active.Import** — Sets the instantaneous power demand in Watts. Negative values request discharge from EV to grid (V2G). Positive values request charge from grid to EV. Zero stops power flow.

These keys are not defined in the OCPP 1.6 specification. They are MEA-specific extensions used for V2G/BPT scheduling and control.

2 Section 9 Results

Summary: 5 PASS 0 FAIL 3 WARN 0 SKIP (8 total)

Item	Test Description	Detail / Evidence	Result
9.0	CSMS connection probe — GetConfiguration HTTP 200	`vSECC connected to MEA CSMS`	**PASS**
9.1–9.3 Configuration (Standard Keys via ChangeConfiguration)			
9.1	ChangeConfiguration V2GMode=true → Accepted	`HTTP 200`	**PASS**
9.2	ChangeConfiguration MeterValueSampleInterval=10 → Accepted	`HTTP 200`	**PASS**
9.3	ChangeConfiguration LocalAuthorizeOffline=false → Accepted	`HTTP 200`	**PASS**
9.4 Power Demand (V2G/BPT) — MEA-Specific Keys			
9.4.0	ChangeConfiguration V2GMode=true (pre-condition for power demand)	`HTTP 200`	**PASS**
9.4.1	Power Demand -5000 W (BPT discharge from EV to grid)	`HTTP 200; metervalues not observed in 15s` *V2G/BPT requires active charging session and V2G-capable EV*	**WARN**

Item	Test Description	Detail / Evidence	Result
9.4.2	Power Demand +2000 W (BPT charge EV from grid)	`HTTP 200; metervalues not observed in 15s` *V2G/BPT requires active charging session and V2G-capable EV*	WARN
9.4.3	Power Demand 0 W (BPT idle / stop power flow)	`HTTP 200; metervalues not observed in 15s` *V2G/BPT requires active charging session and V2G-capable EV*	WARN

3 Notes on Failures and Warnings

9.1 – 9.3 Configuration items

Items 9.1–9.3 call `POST /EV/remote/changeConfiguration` with standard OCPP 1.6 keys (`V2GMode`, `MeterValueSampleInterval`, `LocalAuthorizeOffline`). A result of HTTP 200 from the MEA CSMS confirms the command was forwarded to the charger. If the charger returns `Rejected` in the response body, the detail field will show `status=Rejected`; this indicates the vSECC does not support that key.

9.4 Power Demand (V2G/BPT) items

- **9.4.0 – 9.4.3** use the MEA-specific key `Power.Active.Import` to set an instantaneous power demand. The key is not defined in the OCPP 1.6 specification; it is a MEA extension for BPT (bidirectional power transfer) scheduling.
- **PASS** is assigned when the MEA API returns HTTP 200, indicating the CSMS accepted and forwarded the command.
- **WARN** is assigned when `metervalues` events are not observed on the vSECC MQTT broker within the timeout. This is expected when no active charging session exists or when the connected EV does not support V2G/ISO 15118 -2 power scheduling.
- V2G power control requires: (a) an active charging session, (b) a V2G-capable EV, (c) a signed `PowerDeliveryReq` exchange at the ISO 15118 level. The MEA REST API call alone is not sufficient to verify end-to-end power delivery.

4 Dependency Map

Configuration	9.0 (connection) → 9.1, 9.2, 9.3 (independent)
V2G/BPT	9.4.0 (V2GMode=true) → 9.4.1 → 9.4.2 → 9.4.3